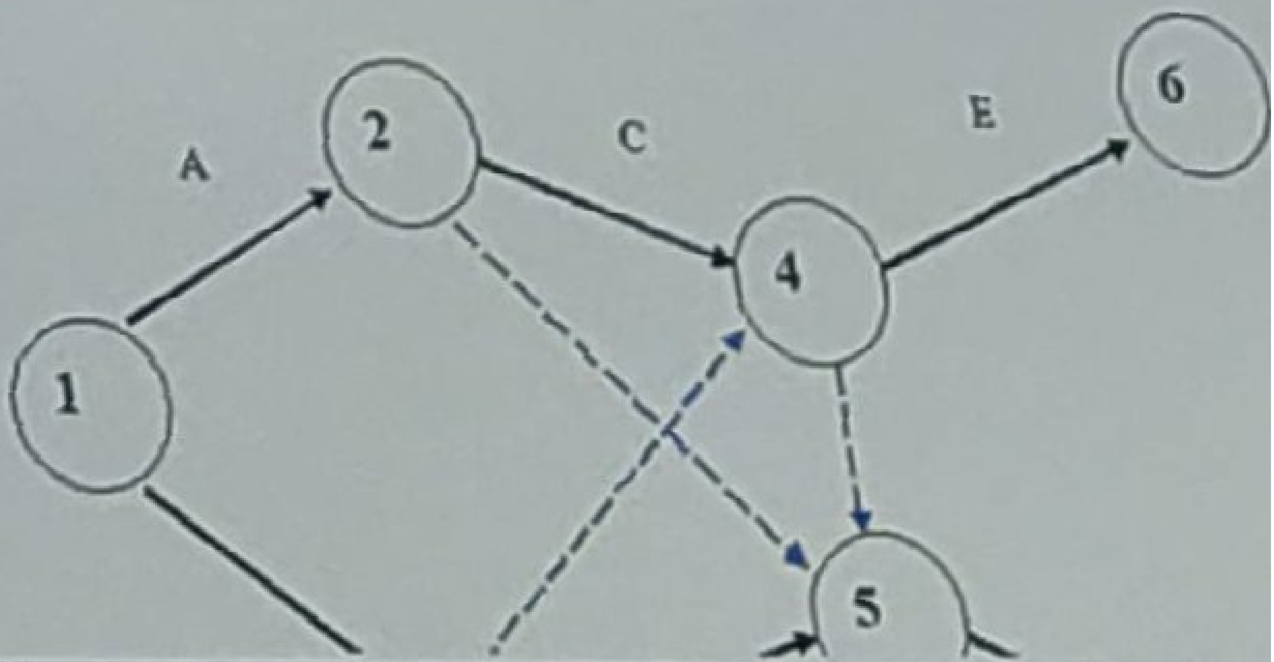
__________ is a network where every pair of nodes is connected

Choices

- Connected network
- Open network
- Closed network
- Unconnected network

Question: 2 (Marks: 1)

Reason for the following network to be illegal?




Choices:

- More than one dummy activity has been introduced
- There does not exist a unique end node
- Dummy activities are intersecting

Question: 3 (Marks: 1)

Critical path is obtained by connecting the jobs having

Choices:

- Activities having same EST and LST
- Activities having same EFT and LFT
- Activities having zero slack
- All of these

Operations Research (MTH601)

Question: 4 (Marks: 1)

If we have 2, 4 and 12 hours as the optimistic time($t_o$), most likely time ($t_m$) and pessimistic time ($t_p$) respectively then the expected time for the activity will be

2

Choices:

- 5 hours
- 4 hours
- 6 hours
- 2 hours

Operations Research (MTH601)

Question: 3 (Marks: 1)

Critical path is obtained by connecting the jobs having

Choices:

- Activities having same EST and LST
- Activities having same EFT and LFT
- Activities having zero slack
- All of these

Operations Research (MTH601)

Question: 3 (Marks: 1)

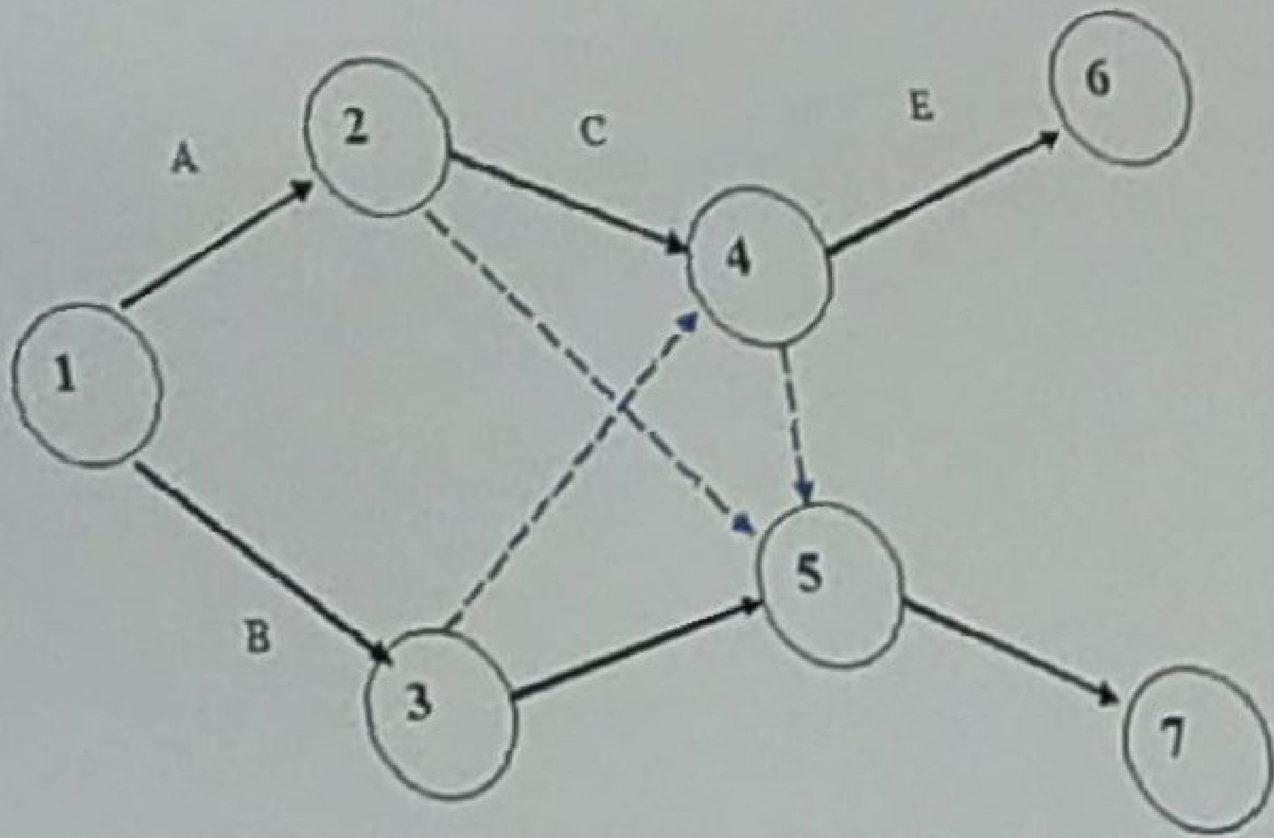
Critical path is obtained by connecting the jobs having

Choices:

- Activities having same EST and LST
- Activities having same EFT and LFT
- Activities having zero slack
- All of these

Question: 2 (Marks: 1)

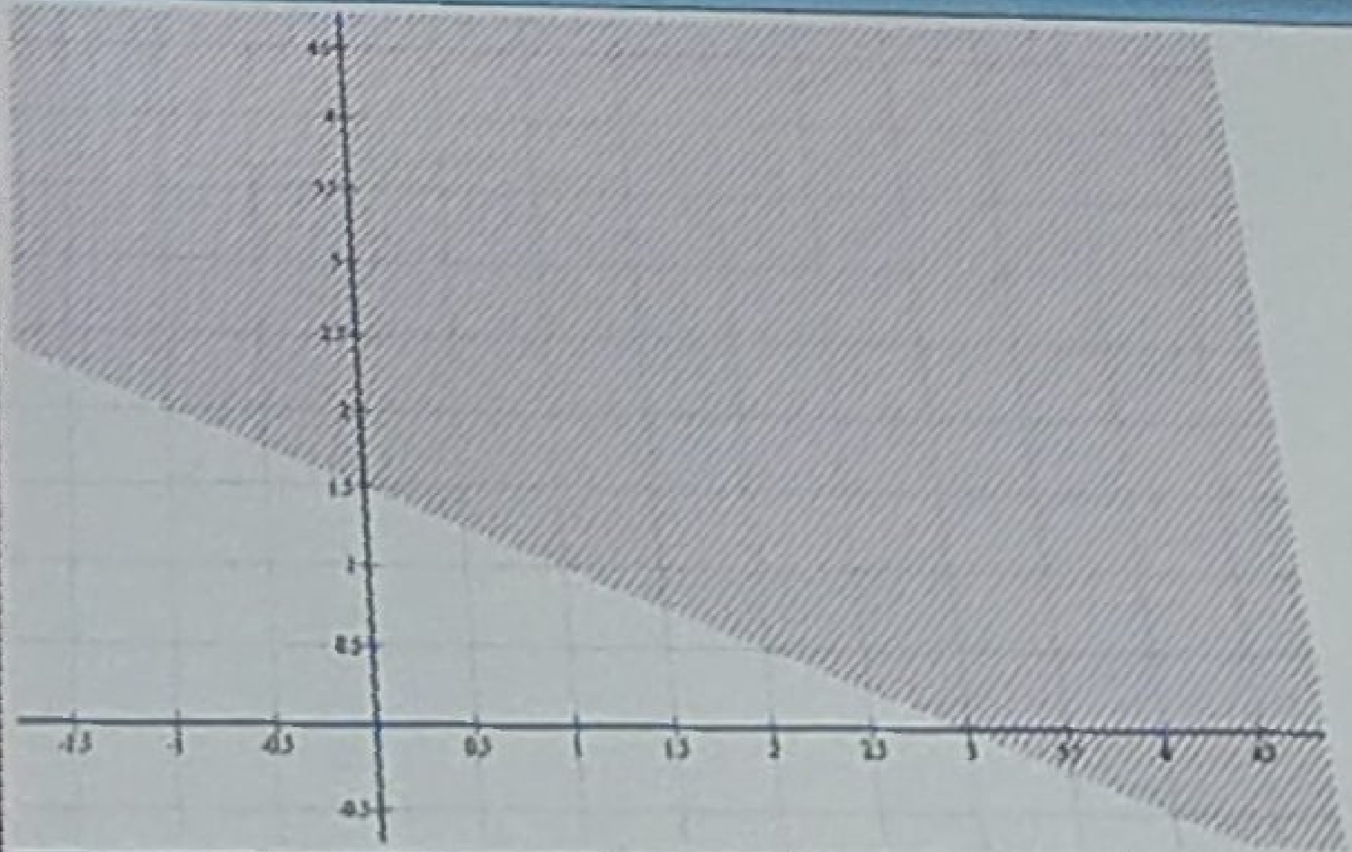
Reason for the following network to be illegal?




Choices:

- More than one dummy activity has been introduced
- There does not exist a unique end node
- Dummy activities are intersecting

**Question: 8** (Marks: 1)



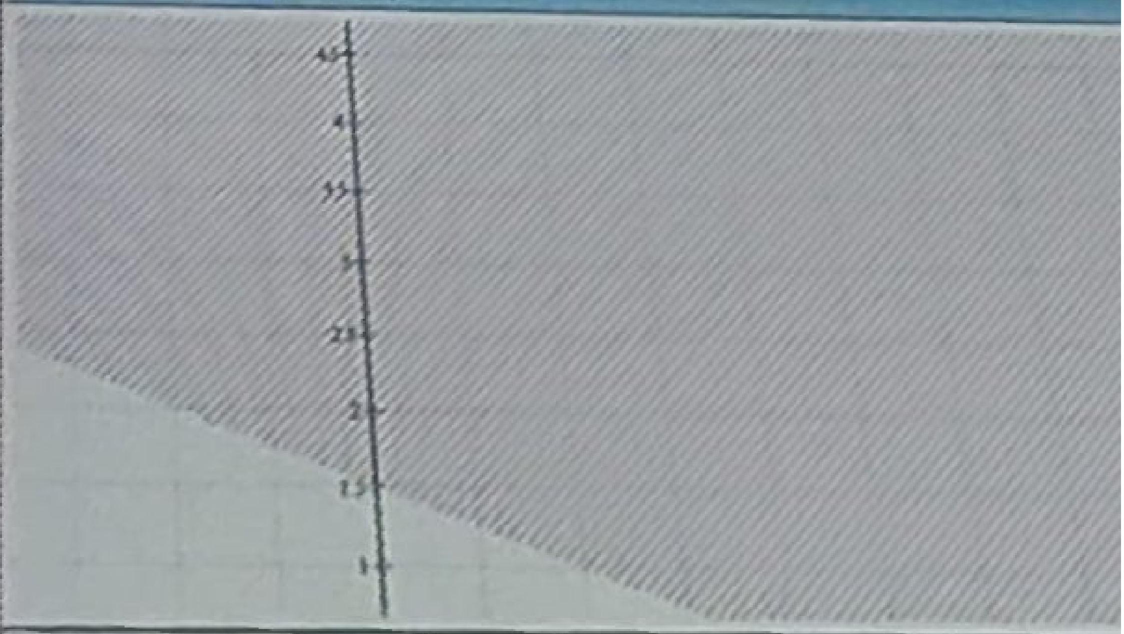
**Choices:**

- $x+2y\leq 3$
- $x+2y\geq 3$
- $x-2y\leq 3$
- $x-2y\geq 3$

TIME LEFT

Operations Research (MTH601)

Question: 8 (Marks: 1)




**Choices:**

- $x+2y\leq 3$
- $x+2y\geq 3$
- $x-2y\leq 3$
- $x-2y\geq 3$

Operations Research (MTH501)

Question: 7 (Marks: 1)

The graphical method for solving a linear programming problem is applicable when ------

Choices:

- two variables are involved
- three variables are involved
- any number of variables are involved
- none of these

Operations Research (MTH601)

Question: 5 (Marks: 1)

If $t_0 = 2$, $t_m = 5$ and $t_p = 14$, then $V_t =$ ________

Choices:

- 2 hrs
- 3 hrs
- 4 hrs
- 5 hrs

Operations Research (MTH601)

Question: 10 (Marks: 1)

If a company manufacture 'x' units of product 'A' and 'y' units of 'B' with associated profits of Rs.5 and Rs.3 then which of the following is the objective function to maximize is the profit?

Choices:

- $z = 15xy$
- $z = 5x - 3y$
- $z = 3x - 5y$
- $z = 5x + 3y$

Operations Research (MTH601)

Question: 9 (Marks: 1)

Which of the following is equivalent relation to the constraint: $x = 2$ of a linear programming problem?

Choices:

- $x \geq 2$
- $x \leq 2$
- $x < 2$ and $x \geq 2$
- $x < 2$ and $x > 2$

may be

Question: 9 (Marks: 1)

Which of the following is equivalent relation to the constraint: $x = 2$ of a linear programming problem?

Choices:

- $x \geq 2$
- $x \leq 2$
- $x < 2$ and $x \geq 2$
- $x < 2$ and $x > 2$

Operations Research (MTH601)

Question: 6 (Marks: 1)

One of the properties of Linear Programming Model is--------

Choices:

- It will not have constraints
- It should be easy to solve
- It must be able to adopt to solve any type of problem
- The relationship between problem variables and constraints must be linear

Operations Research (MTH601)

Question: 13 (Marks: 1)

In the simplex table for a linear programming problem, we select the leaving basic variable corresponds to ----------

Choices:

- Maximum non-negative ratio
- Maximum negative ratio
- Minimum non-negative ratio
- Minimum negative ratio

Operations Research (MTH601)

Question: 12 (Marks: 1)

If the objective function of a linear programming problem needs further improvement then which of the following will have to proceed?

Choices:

- a new decision variable to be entered and other decision variable to leave the basis
- a new slack variable to be entered and other slack to leave the basis
- a new non-degenerate variable to be entered and other non-generate to leave the basis
- a new basic variable to be included and other basic variable to leave the basis

may be

Question: 11 (Marks: 1)

While using Simplex method for a LPP, if there are 4 standard equations in 8 variables, then the number of non-basic variables will be ——

Choices:

- 32
- 12
- 4
- 2

Question: 17 (Marks: 1)

Which of the following rule in used in order to minimize the decision variables in any Inventory model?

Choices:

- Derivative test
- Simplex method
- PERT
- Graphical method

...tion: 15 (Marks: 1)

_______ is any stored resource that is used to satisfy a current or future need.

Choices:

- Inventory
- Simulation
- Allocation
- Assignment

Operations Research (MTH601)

Question: 16 (Marks: 1)

For an inventory item if the demand rate per year and the economic order quantity 10,000 and 912.87 respectively. Then the number of orders per year are ———.

$$N = \frac{D}{Q}$$

Choices:

- 10.95 ✓
- 13.65
- 1.44
- 4.50

TIME LEFT

Operations Research (MTH601)

Question: 14 (Marks: 1)

While solving a linear programming problem by using Simplex method, the key row indicates ——

Choices:

- artificial variable
- slack variable
- leaving variable
- entering variable

TIME LEFT

Question: 18 (Marks: 1)

Under which of the following condition, we have no holding and setup cost in the Manufacturing Model with no shortage?

Choices:

- Demand rate > Production rate
- Demand rate < Production rate
- Demand rate = Production rate
- Production rate $\to 0$ or Production rate $\to \infty$

Question: 19 (Marks: 1)

The basis for ABC analysis is-------

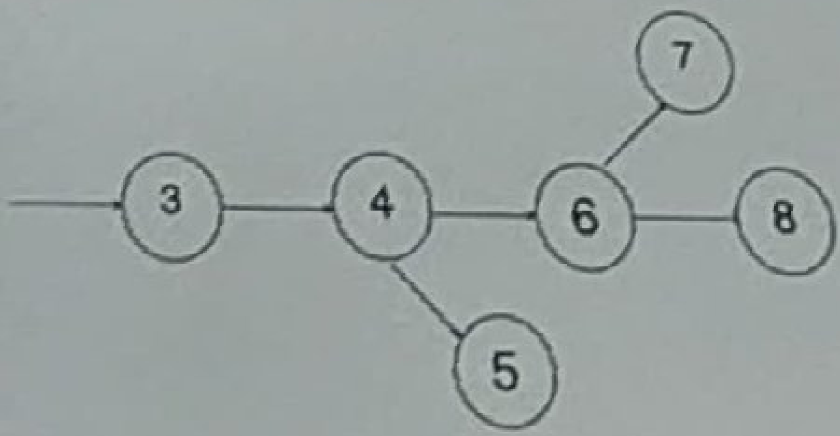
Choices:

- Wilson Model
- Manufacturing Model without shortage
- Manufacturing Model with allowed shortage
- Pareto's 80-20 rule (80 % countries economy is controlled by 20% of people)

TIME LEFT

Question: 20 (Marks: 1)

The given network diagram is an example of ________.




Choices:

- Looping
- Merge node
- Burst node
- Dangling

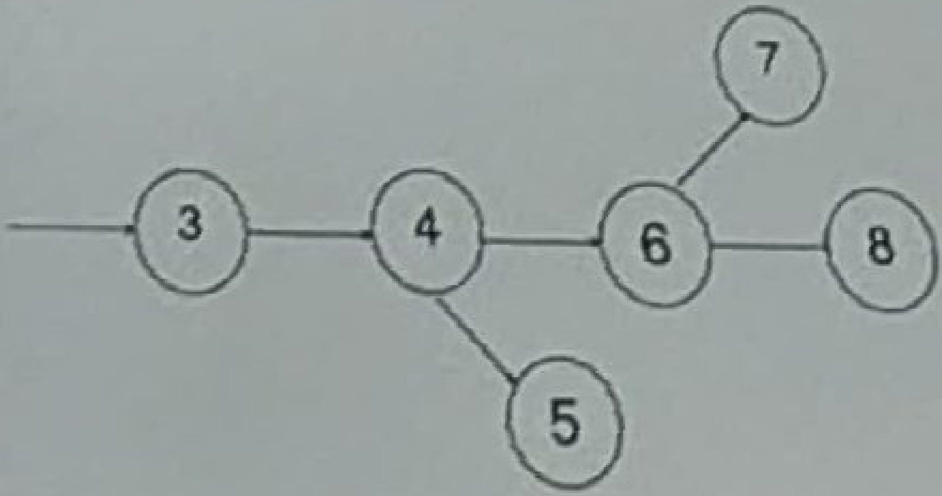
TIME LEFT

56

Search

Question: 20 (Marks: 1)

The given network diagram is an example of________.




Choices:

- Looping
- Merge node
- Burst node
- Dangling

Operations Research (MTH601)

Question: 21 (Marks: 1)

Which one of the following is not an operations research problem solving step?

Choices:

- Observation
- Definition of the problem
- Variable selection
- Model construction

Operations Research (MTH601)

Question: 26 (Marks: 1)

Minimizing, $z = \sum_{i=1}^{n} c_i x_i$ is equivalent to...

Choices:

- Maximize, $-z = \sum_{i=1}^{n} c_i x_i$
- Maximize, $-z = \sum_{i=1}^{n} (-c_i) x_i$
- Maximize, $z = \sum_{i=1}^{n} -c_i x_i$
- Maximize, $z = \sum_{i=1}^{n} c_i x_i$

Operations Research (MTH601)

Question: 24 (Marks: 1)

............................ is used for calculating the latest occurrence time.

Choices:

Forward pass

Backward pass

Operations Research (MTH601)

Question: 23 (Marks: 1)

Looping is allowed in a network diagram.

Choices:

- True
- False

Operations Research (MTH601)

Question: 22 (Marks: 1)

Game theory problems are covered by which of the following operations research techniques?

Choices:

- Linear programming techniques
- Network techniques

maybe

- Inventory techniques
- Probabilistic techniques

Operations Research (MTH601)

Question: 25 (Marks: 1)

$x \le 0$ and $y \ge 0$ shows that the point lies in-----

Choices:

- 1st quadrant
- 2nd quadrant
- 3rd quadrant
- 4th quadrant

Operations Research (MTH601)

Question: 27 (Marks: 1)

Under which of the following substitution, the econmic order quantity:

$$Q^* = \sqrt{\frac{2C_2D}{C_3\left(1-\frac{D}{R}\right)}}\sqrt{\frac{C_3+C_4}{C_4}}$$

for the Manufacturing Model with shoratge can be reduced into Wilson Model?

Choices:

- $R \to 0, C_4 \to 0$
- $R \to \infty, C_4 \to \infty$
- $R \to 0, C_4 \to \infty$
- $R \to \infty, C_4 \to 0$

Operations Research (MTH601)

Question: 28 (Marks: 3)

Write the standard form of following linear programming problem;

$Max\ Z = 5x + 4y$

*Subject to:*

$$
\begin{aligned}
6x + 4y &\le 24 \\
x + 2y &\le 6 \\
-x + y &\le 1 \\
y &\le 2 \\
x \ge 0,\ y &\ge 0
\end{aligned}
$$

Answer:

TIME LEFT

56

Search

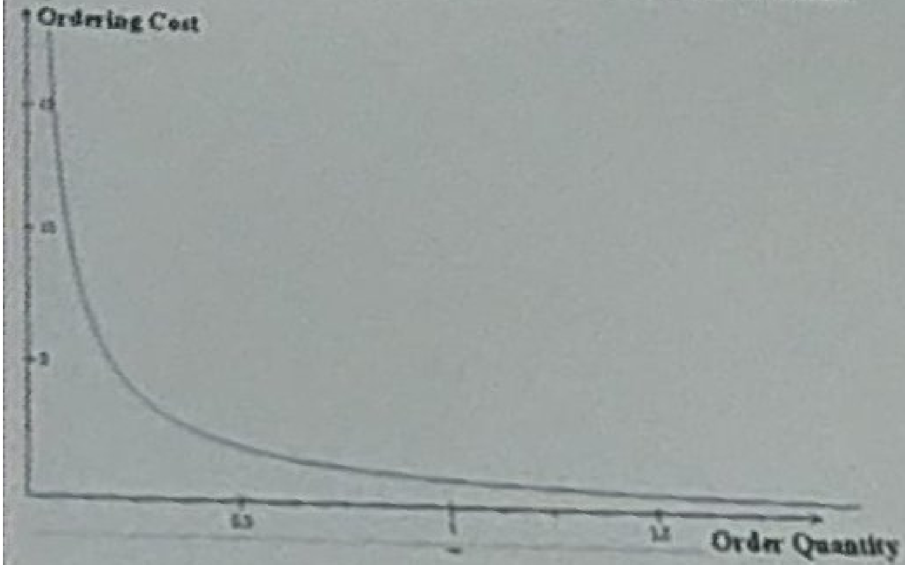
Question: 29 (Marks: 3)

For the following graph:

(i) Discuss the proportionality relation b/w the inventory terms.

(ii) What the slope of curve represents?

(iii) When does the ordering cost per unit item would be minimum?




Answer:

TIME LEFT

55

Search

Operations Research (MTH601)

Question: 30 (Marks: 5)

Express the objective function of the following linear programming problem in terms of non-basic variables and hence evaluate the initial basic feasible solution.

Minimize $Z = 9x_1 + x_2$

*Subject to:*

$4x_1 + 3x_2 = 5$

$7x_1 + 5x_2 \geq 3$

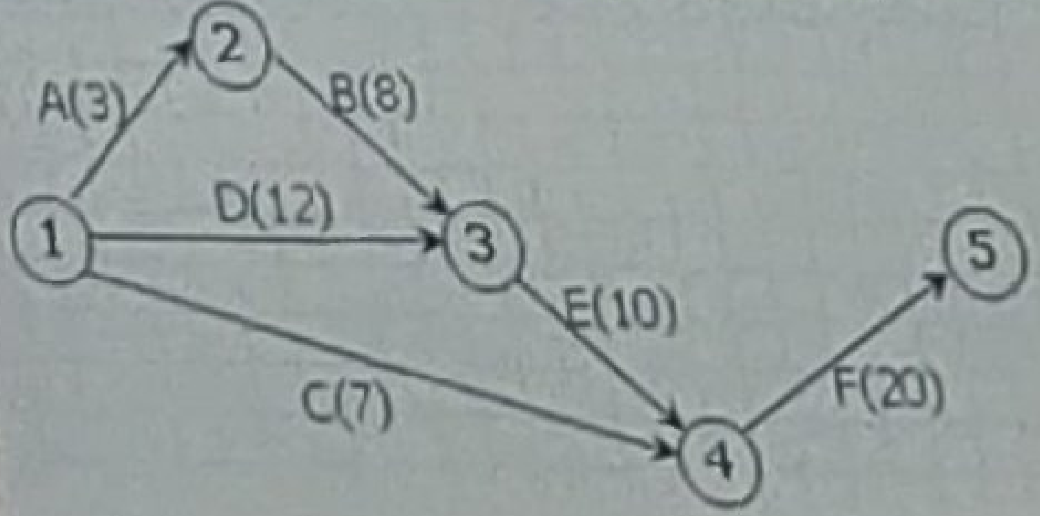
$2x_1 + 4x_2 \leq 7$

$x_1, x_2 \geq 0$

Answer:

...ations Research (MTH601)

...estion: 31 (Marks: 5)




Construct a table that shows predecessors of activities for the above network diagram.

Operations Research (MTH601)

**Question: 1 (Marks: 1)**

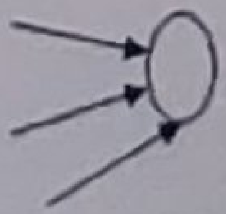
The dummy activities consume

**Choices:**

- No time, no resources
- No time but some resources
- Some resources in minimum time
- All resources in a given time

Question: 2 (Marks: 1)

The following event is an example of



Choices:

- Dangling
- Bursting
- Merging
- Dummy

perations Research (MTH501)

Question: 3 (Marks: 1)

For any activity backward pass computations provide its

Choices:

Earliest start times

Latest start times

Moderate start times

Completion time

Operations Research (MTH601)

Question: 3 (Marks: 1)

For any activity backward pass computations provide its

Choices:

- Earliest start times
- Latest start times
- Moderate start times
- Completion time

perations Research (MTH601)

Question: 4 (Marks: 1)

Critical path is obtained by connecting the jobs having

Choices:

- Activities having same EST and LST
- Activities having same EFT and LFT
- Activities having zero slack
- All of these

Operations Research (MTH601)

Question: 9 (Marks: 1)

Solution region of the constraint $x \geq 0$ is------

Choices:

- half plane to the right of straight line $x = 0$
- half plane to the right of y-axis
- half plane to the region where abscissas are non-negative
- all are equivalent

Operations Research (MTH601)

Question: 10 (Marks: 1)

Which of the following is equivalent relation to the constraint: $x = 2$ of a linear programming problem?

Choices:

- $x \geq 2$
- $x \leq 2$
- $x < 2$ and $x \geq 2$
- $x < 2$ and $x > 2$

may be

Operations Research (MTH601)

Question: 11 (Marks: 1)

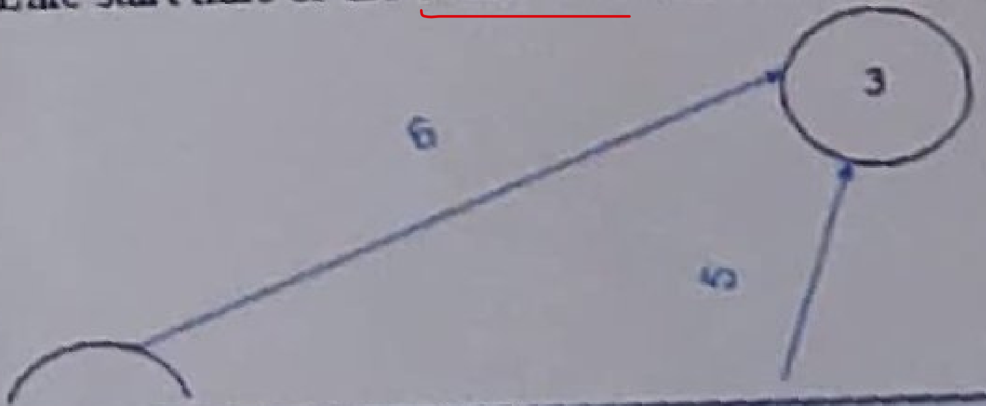
At any stage if a manufacturing company of products 'A' and 'B' stops the production or manufacture any amounts say $x$ and $y$ respectively of these products then which of the following are the valid constraints associated with this stage of production?

Choices:

- $x \geq 0, y \geq 0$
- $x > 0, y > 0$
- $x \leq 0, y \leq 0$
- $x < 0, y < 0$

**Question: 5 (Marks: 1)**

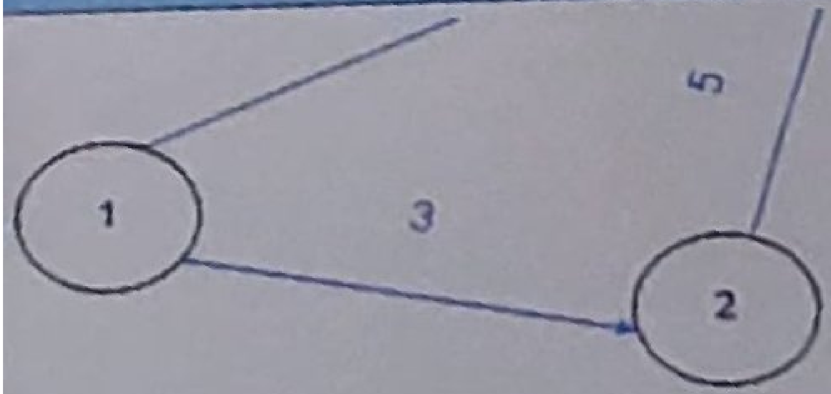
Late start time of the event '1' of the following project is---------




**Choices:**

- Zero day
- 1 day
- 2 days
- 3 days

Question: 5 (Marks: 1)




Choices:

- Zero day
- 1 day
- 2 days
- 3 days

Operations Research (MTH601)

Question: 6 (Marks: 1)

If $t_o = 10$, $t_m = 10$ and $t_p = 16$, then $S.D =$ ________.

Choices:

- 11
- 121
- 1
- 12

Operations Research (MTH601)

Question: 7 (Marks: 1)

Which of the following is an essential characteristic of a linear programming problem?

Choices:

- The relationship between variables and constraints must be linear
- The model must have an objective function
- The model must have structural constraints
- All are equivalent

Operations Research (MTH601)

Question: 8 (Marks: 1)

In a linear Programming Problem (LPP), which of the following must be hold?

Choices:

- Only objective function is linear
- Both objective function and constraints are linear
- Only constraints needs to be linear
- At least one of objective function or constraint should be linear

Operations Research (MTH601)

Question: **12** (Marks: 1)

While using Simplex method for a LPP, if there are 4 standard equations in 8 variables, then the number of non-basic variables will be ------

Choices:

- 32
- 12
- 4
- 2

Operations Research (MTH601)

Question: 13 (Marks: 1)

Which of the following is the standard equation to the constraint; $x \leq 8$ subject to a linear programming problem?

Choices:

- $x - s = 8$, $s \leq 0$
- $x + s = 8$, $s \leq 0$
- $x - s = 8$, $s \geq 0$
- $x + s = 8$, $s \geq 0$

Operations Research (MTH601)

Question: 14 (Marks: 1)

To convert ≤ type of inequality into equations, we have to ——

Choices:

- add slack variables
- subtract slack variables
- add artificial variables
- assume them to be equations

Operations Research (MTH601)

Question: 12 (Marks: 1)

While using Simplex method for a LPP, if there are 4 standard equations in 8 variables, then the number of non-basic variables will be ------

Choices:

- 32
- 12
- 4
- 2

Operations Research (MTH601)

Question: 16 (Marks: 1)

If an inventory is made by taking the loan from the bank and we pay the interest at the rate say $x\%$ annually, then this interest is the main part of --------

Choices:

- Item Cost
- Ordering Cost
- Inventory holding cost
- Shortage Cost

Operations Research (MTH601)

Question: 17 (Marks: 1)

If the volume of inventory increases, then which of the following cost will increase?

Choices:

- Stock out cost
- Ordering cost
- Item cost
- Inventory carrying cost

Operations Research (MTH601)

Question: 18 (Marks: 1)

Which of the following rule is used in order to minimize the decision variables in any Inventory model?

Choices:

- Derivative test
- Simplex method
- PERT
- Graphical method

Operations Research (MTH601)

Question: 20 (Marks: 1)

Which of the following cost remains almost unchanged in both Manufacturing and Purchasing Models without shortages?

Choices:

- Total Inventory Cost
- Item Cost
- Holding Cost
- Setup Cost

Operations Research (MTH601)

Question: 22 (Marks: 1) Attempte

In non-linear programming either the objective function or the constraints or both can be

---------------

Choices:

- Continuous functions
- Non-linear functions
- Linear functions
- Constant functions

Operations Research (MTH601)

Question: 23 (Marks: 1)

Looping is allowed in a network diagram

Choices:

True

False

Operations Research (MTH601)

Question: 24 (Marks: 1) Ans

A forward pass is used to determine and calculate the ____________ dates, through utilization of a previously specified start date

Choices:

- early start and late finish
- early start and early finish
- late start and late finish
- late start and early finish

Operations Research (MTH601)

Question: 25 (Marks: 1)

$x \leq 0$ and $y \geq 0$ shows that the point lies in-----

Choices:

- 1st quadrant
- 2nd quadrant
- 3rd quadrant
- 4th quadrant

Operations Research (MTH601)

Question: 28 (Marks: 3)

The demand for an item is 15000 units / year, cost of one purchase is Rs. 200 and the holding cost is Rs. 1 per unit per year. If the shortage cost is Rs. 5 / year / item Then determine the Optimum Order Quantity.

Answer:

Question: 27 (Marks: 1)

Under which of the following substitution , the econmic order quantity:

$$Q^* = \sqrt{\frac{2C_2D}{C_3\left(1-\frac{D}{R}\right)}}\sqrt{\frac{C_3+C_4}{C_4}}$$

for the Manufacturing Model with shoratge can be reduced into the same Model without shortage?

Choices:

- $C_3 \to \infty$
- $C_3 \to 0$
- $C_4 \to \infty$
- $C_4 \to 0$

Question: 26 (Marks: 1)

Minimizing, $z = \sum_{i=1}^{n} c_i x_i$ is equivalent to...

Choices:

- Maximize, $-z = \sum_{i=1}^{n} c_i x_i$
- Maximize, $-z = \sum_{i=1}^{n} (-c_i) x_i$
- Maximize, $z = \sum_{i=1}^{n} -c_i x_i$
- Maximize, $z = \sum_{i=1}^{n} c_i x_i$

**Question: 21 (Marks: 1)**

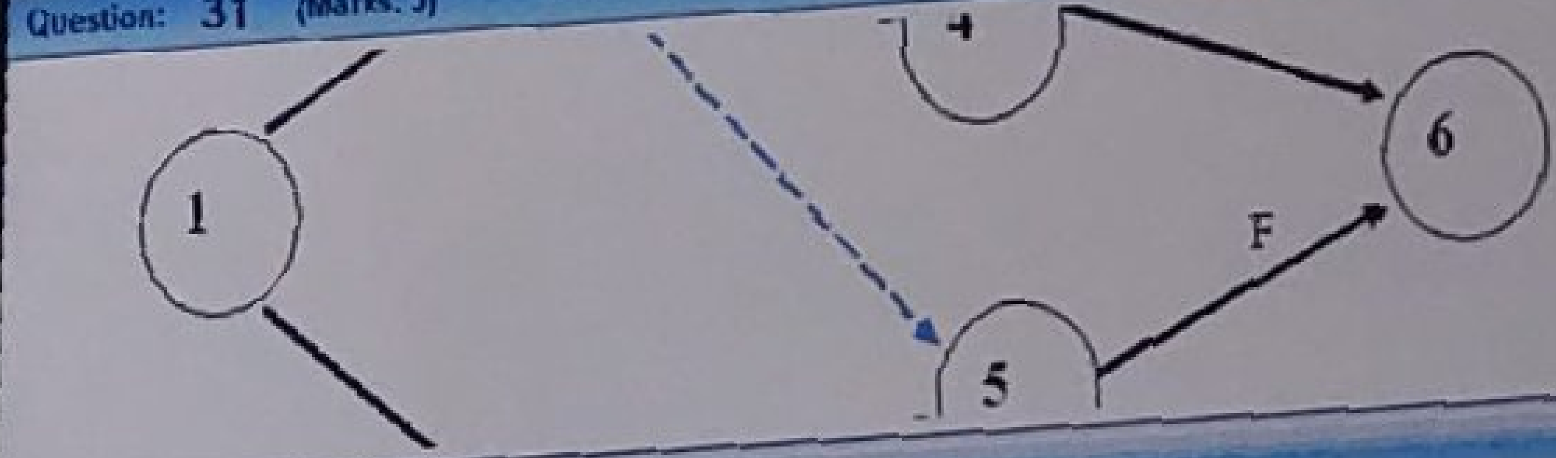
Which one of the following is not an operations research problem solving step?

**Choices:**

- Observation
- Definition of the problem
- Variable selection
- Model construction

Operations Research (MTH601)

Question: 31 (Marks: 5)

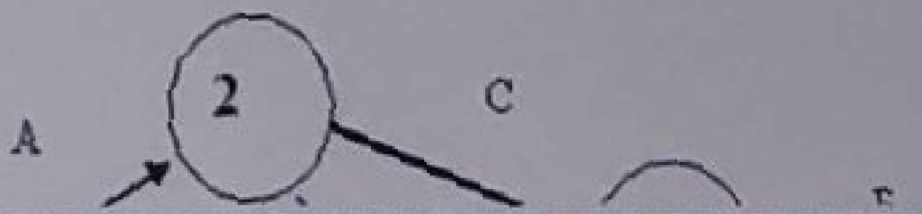



Answer:

Operations Research (MTH601)

Question: 31 (Marks: 5)

Construct a table that shows predecessors of activities in below network diagram.




Answer:

Operations Research (MTH601)

Question: 30 (Marks: 5)

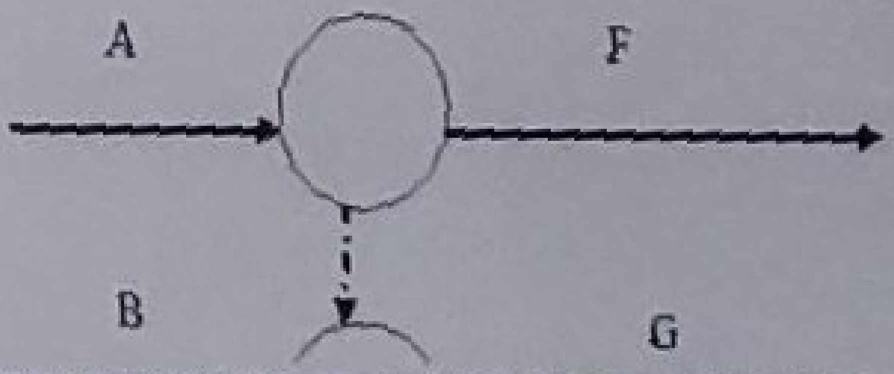
If an inventory model can be represented graphically as follows, then what assumptions can be made about its:

(i) Demand
(ii) Shortage
(iii) Rate of replenishment
(iv) Lead Time (reorder gap)
(v) Lot size for each cycle ?

Answer:

Operations Research (MTH601)

Question: 29 (Marks: 3)




Answer:

# Floats/ Slack

$$I = j_K - i_L - D$$
$$F = j_E - i_E - D$$
$$T = j_L - i_E - D$$

## Latest Finish Time

Latest Event Time of the Head Event

Earliest Time of Tail Event

$$i = iE$$

Latest Time of Tail Event

$$i = iL$$

Earliest Time to Head Event

$$j = jE$$

Latest Time of Head Event

$$j = jL$$

## Earliest start time

Earliest possible time at which an activity can start and is given by the earliest time of the Tail Event.

## Latest Start Time

Latest possible time by which an activity start and is given by subtracting the duration time from the Latest Finish time

## Earliest Finish Time

Earliest possible time at which an activity can Finished and is given by adding the duration time to the earliest start time.

## Latest Finish Time

Latest Event Time of the Head Event

Earliest Time of Tail Event

Earliest possible time at which an activity can start and is given by the earliest time of the Tail Event.

## Latest Start Time

Latest possible time by which an activity start and is given by subtracting the duration time from the Latest Finish time

## Earliest Finish Time

Earliest possible time at which an activity can Finished and is given by adding the duration time to the earliest start time.

## Latest Finish Time

Latest Event Time of the Head Event

Earliest Time of Tail Event

$$i = iE$$

$$I = jE - iL - D$$

If negative

Take I=0

## Free Float

Time by which an activity can expand without effecting subsequent activity.

$$F = jE - iE - D$$

## Total Float

Time by which an activity can expand without effecting the overall duration of the project.

$$T = jL - iE - D$$

**Target Time > Project Time**

- **→ Positive Float**
- **Target Time < Project Time**
- **→Negative Float**
- **→Reduce the Associated Activity**

## Independent Float

Time by which an activity can expand without affecting other

(Prev or subseq)

$$I = jE - iL - D$$

If negative

Take I=0

## Free Float

Time by which an activity can expand without effecting subsequent activity.

$$F = jE - iE - D$$

## Total Float

Time by which an activity can expand without effecting the overall duration of the project.

$$T = jL - iE - D$$

Target Time > Project Time